STATISTICS

Chapter 2 Homework

Describing distributions

Sections 2.1-2.7 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Displays, position, and shape

1. The following are waiting times in minutes: 2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 8, 19. Make a stem-and-leaf plot with a key. Identify a gap and describe the tail direction.

2. Use the data in Problem 1 to make a frequency histogram with intervals [0, 5), [5, 10), [10, 15), [15, 20). Label axes and explain where a wait of exactly 5 belongs.

3. Find the five-number summary and IQR for Problem 1. Use the median-of-halves convention. Find the 1.5-IQR fences and identify any outlier.

4. Draw a modified box plot for Problem 1: whiskers end at the most extreme non-outliers, and outliers are plotted separately. Does a longer quarter of a box plot necessarily contain more observations?

5. Find the mean and median for Problem 1. Which better represents a typical wait here, and why? Predict how each changes if 19 is replaced by 39, then check.

Spread, weights, and choice of display

6. For the sample 2, 4, 4, 6 minutes, make columns for each value, deviation from the mean, and squared deviation. Find the sample variance and sample SD, with units. You may use $\sqrt{(8/3)} \approx 1.633$. Why divide by 3 instead of 4?

7. Two groups have the same mean of 10 minutes; their SDs are 2 and 5 minutes. In which group is a 14-minute wait more unusual in SD units? Compute both standardized distances; do not assume normality.

8. A class of 10 students has mean 12 points; another class of 30 has mean 16 points. Find the combined mean. Explain why simply averaging 12 and 16 is wrong.

9. Choose a bar chart, histogram, or time-series graph for (a) counts by course subject, (b) distribution of individual wait lengths, and (c) average wait each day for a month. Explain one choice.

10. Every waiting time in a dataset increases by 3 minutes. What happens to its mean, median, IQR, and SD? Explain without recomputing the dataset.

11. A distribution is approximately bell-shaped with mean 20 and SD 3. Use the empirical rule to estimate the fraction between 14 and 26. Could you make the same estimate from mean and SD alone for a strongly skewed distribution?